(a) steering $f\mathfrak{A}^i/\delta_E$

$f\mathfrak{A}^i/\delta_E$

10^0

10^{-5}

exact
exact, Ek/10
inviscid H/f
deep: $(H-\delta_E/2)/f$
shallow: $(8/15)\,s^5$

10^{-1} 10^0 10^1

$s = H/\delta_E$

(b) friction $-f\mathfrak{A}^r/\delta_E$

$-f\mathfrak{A}^r/\delta_E$

10^0 10^{-1} 10^{-2} 10^{-3} 10^{-4}

exact
exact, Ek/10
deep: $\delta_E/(2f)$
shallow: $(2/3)\,s^3$

10^{-1} 10^0 10^1

$s = H/\delta_E$

(c) steering/friction $\mathfrak{A}^i/(-\mathfrak{A}^r)$

$\mathfrak{A}^i/(-\mathfrak{A}^r)$

10^0 10^{-2}

exact
exact, Ek/10
deep: $2s-1$
shallow: $(4/5)\,s^2$

10^{-1} 10^0 10^1

$s = H/\delta_E$

(d) bottom stress $\mathrm{Ek}|S|/\delta_E$

$\mathrm{Ek}|S|/\delta_E$

10^0 10^{-1}

exact
exact, Ek/10
deep: $1/\sqrt{2}$
shallow: s

10^{-1} 10^0 10^1

$s = H/\delta_E$

(e) bottom-stress angle $\arg S$

degrees

10 0 −10 −20 −30 −40 −50 −60

exact
exact, Ek/10
deep: -45°
shallow: 0°

10^{-1} 10^0 10^1

$s = H/\delta_E$

(f) surface stress at the bed

$|\operatorname{sech}\varphi H|$

10^0 10^{-5}

exact
exact, Ek/10
deep: $2e^{-s}$

10^{-1} 10^0 10^1

$s = H/\delta_E$